ANTHONY D. CARPENETTI

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DESIGN ENGINEER — PREFERRED NAME: TONY

Mechanical Design Engineer with 9+ years of experience delivering solutions across aerospace, automotive, and manufacturing industries. Currently supporting NASA's Space Launch System (SLS) Artemis Program as a vehicle integration engineer with direct experience producing flight-critical interface documentation and analyses on the ESSCA Contract. Proven track record combining technical expertise with strong communication skills and a self-driven approach to continuous learning and improvement.

Core competencies: Mechanical Design, CAD Modeling and Simulation, GD&T, FEA, Technical Writing and Presentation, Product Design, Project Management, Communication, Code Writing & Debugging, Data Analysis, Critical Thinking, and Public Speaking.

CAREER HIGHLIGHTS

- Supported successful 99.92% Artemis II Insertion Accuracy resulting from design and tolerance analysis of flight hardware
- Received the August 2022 Outstanding Task Award from Amentum for successful accelerated delivery of Mechanical Interface Control Drawings (MICDs) for the Block 1B Exploration Upper Stage (EUS) Forward and Aft Umbilical Interfaces supporting Vehicle Integration on NASA's SLS Artemis II Program

PROFESSIONAL EXPERIENCE

BEVILACQUA RESEARCH CORPORATION (BRC) Huntsville, AL — (40hrs/week) **FEB. 2021 – PRESENT**
[**NASA's ESSCA Contract** with **Amentum Space Exploration Division (ASED)** at **Marshall Space Flight Center (MSFC)**]
Mechanical Design Engineer, SLS Artemis Program (EV32 – Structural Design, Development, and Analysis)

- Integrate large assemblies and subassemblies with interfaces and envelopes, working both individually and with a team of engineers and designers to develop CAD models, evaluate hardware components and systems, and advance NASA's missions
- Provide engineering design support for insight and oversight of time-critical hardware design cycles and flight/ground interfaces
- Delivered reviews for numerous Block 1, Block 1B, and Block 2 flight configuration projects, documenting and correcting Booster Nozzle rotational errors, reference designator misalignments, broken geometries in lightweight viewables and CAD models and various instances of inconsistent data reporting. Assured customer satisfaction through iterative reviews and comment tracking
- Led a month long review cycle for flight-critical static tolerance envelope drawings, performing overlay comparisons and tracking comments to ensure 100% design accuracy to customer requirements and NASA standards
- Improved the data validation and acceptance process as measured by 50% reduction in required time and significant removal of human error by creating an automated Excel Query for pulling new data and populating the associated tolerance stacks
- Redesigned an Engineering Parts List (EPL) template with improved formatting and increased visibility, accuracy, and usability

REGAL REXNORD Tipp City, OH — (40hrs/week) **NOV. 2018 – FEB. 2021**
Mechanical Design Engineer

- Led the design and production of a motor frame retrofit which avoided new tooling and reduced project costs by $20,000
- Drove the innovation of a novel hall sensor mounting solution to overcome space constraints within a motor housing
- Managed ball bearing project from conceptual design through CAD modeling and simulation, benchmarking, testing, supplier quoting, manufacturing, and customer sign-off. Met the need for a sturdier motor at the same frame size and price bracket
- Supported $37.61M cost-out savings with a team using Value Analysis and Value Engineering (VAVE) and 80/20 methodology

HITACHI ASTEMO Mt. Sterling, OH — (40hrs/week) **SEPT. 2017 – NOV. 2018**
Project Engineer

- Led lifecycle management, quality, and delivery on new product releases for a tier-1 OEM automotive manufacturing supplier
- Reduced evaporator core assembly time by 20% via the implementation of a robotic assembly arm. Programmed the Human Machine Interface (HMI) to reduce the initial scrap rate and achieve 97% good part throughput
- Designed process improvement solutions, developed assembly support fixtures and jigs, managed the installation and repair of production line equipment, and streamlined part flow paths for increased ergonomics and efficiency

BWI GROUP Dayton, OH | **BASIC SYSTEMS, INC** Cambridge, OH — (Seasonal 40hrs/week) **MAY 2014 – MAY 2017**
Mechanical Engineering Intern

- Designed and tested magnetorheological fluid automotive suspension systems, analyzing 1 million+ cycles of durability test data
- Diagnosed a manufacturing cross-contamination issue and presented an intermediate parts-washing process solution
- Fostered international and cross-cultural teamwork and communication with Mexican production facilities and coworkers
- Produced technical specifications, Process Failure Mode and Effects Analyses (PFMEAs), Process Flow Diagrams (PFDs), Process Control Plans (PCPs), & quotes. Honed CAD drafting skills while adhering to ASME Y14.5 standards and procedures

SKILLS AND TOOLS

Programming & Web Dev Software: Git, GitHub, HTML, CSS, JavaScript, Typescript, Python, Visual Basic, SQL, C++, C
Design Tools: CREO, SolidWorks, ANSYS FEA, AutoCAD, LabVIEW, MATLAB & Simulink, Windchill PLM, Mathcad, CNC
Creative: Affinity (Designer, Photo, Publisher), Digital Audio Workstation (Ableton Live, FL Studio), Adobe, Canva, 3D Printing
Miscellaneous: Microsoft Office Suite (Word, Excel, PowerPoint, Project, Outlook, OneNote, Teams), Scrum, Agile, Kanban

EDUCATION

WRIGHT STATE UNIVERSITY Dayton, OH 45435 **AUG. 2013 – MAY 2017**
B.S. Mechanical Engineering | *Cum Laude - Honors Diploma, GPA: 3.53*

COMMUNITY LEADERSHIP

STEM Advocate – Delivered Mechanical Engineering career presentation to 100s of students at Alma Mater HS Career Expo. Member of Amentum Careers Network (ACN), ASED Mentor Peers Program (MPP), Marshall Early Career Organization (MECO)
Local Church Leader – Mentor young adults, prepare weekly study content, develop digital content, play on the worship team
Technology Consultant – Serve as an informal technology advisor for an extended network of family and friends: Diagnose hardware, software, and connectivity issues while translating complex technical solutions into easily digestible language. Apply creative and technical skills across personal and professional projects in web development, graphic design, and digital media – examples available at **https://www.tonycarpenetti.com/portfolio**.

OTHER ACHIEVEMENTS

Certificate of Completion for the following:

- Tolerance Stack Analysis for both coordinate and geometric tolerances **SEPT 2025**
- Jacobs (now Amentum) Technical Excellence Development Program **OCT. 2023 – JUNE 2024**
- Advanced Concepts of GD&T based on ASME Y14.5-2009 Standard **JUNE 2023**
- Fundamentals of GD&T based on ASME Y14.5-2009 Standard **AUG. 2022**
- Fundamentals of Geometric Dimensioning and Tolerancing (GD&T) **MAR. 2021**